resembling the ideal expected pattern observed when comparing a model against itself. Ideally, the minimum similarity value should be observed where the layers match. RTD fails to achieve this pattern in the last two layers, while CKA fails in layer five.

4.1.2 Sensitivity Test

An effective structural similarity index should be robust to the removal of low variance principal components and sensitive to the removal of high variance principal components. Previous studies have shown that CKA is insensitive to the removal of principal components until the most significant ones are removed. We conducted a sensitivity test with our ResNet-18 models trained on CIFAR-10, following a similar experimental setup to that used by Ding et al. (2021).

For each index, we set the dissimilarity score to be above the detectable threshold if it exceeds the dissimilarity score between representations from different random initializations. The results on the first and last layer, shown in Figure 1, demonstrate that NSA is more sensitive to the removal of high variance principal components compared to RTD and CKA, with an average detection threshold of 90.25%, compared to 96% for RTD and 99% for CKA.

4.2 NSA as a Loss Function

The viability of NSA as a loss function is established by demonstrating its non-negativity, nullity and continuity, and by developing a differentiation scheme. Proofs for all four properties are presented in Appendix C and F. We also demonstrate that the expectation of NSA over a minibatch is equal to the NSA of the whole dataset, cementing its feasibility as a loss function in Appendix G.

4.2.1 Convergence of Global NSA in mini batching

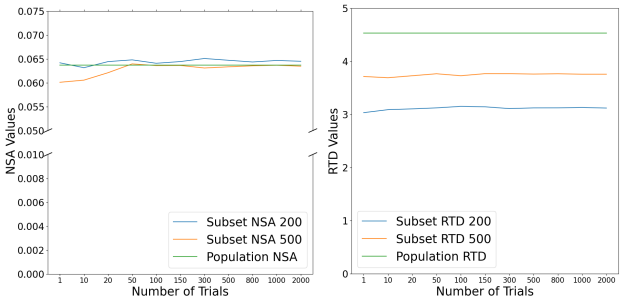
Neural networks are often trained on mini batches, which means that at any step a global structure preserving loss will only have access to a subset of the data. It is vital that a structure preserving loss function like GNSA converge to its global measure when used as a loss function. Similarity indices usually have a high computation cost, therefore most similarity indices work with a sample of the data and show that the value computed on the sample is a good approximation of the global value. Additionally, for a loss function, the expectation is that the mean of the losses from the mini-batches converges to the true loss over the entire dataset. This principle underlies stochastic gradient descent (SGD) and its variants, which are foundational to training neural networks. SGD assumes that the expected value of the gradients calculated on the mini-batches converges to the true gradient over the entire dataset, and a similar concept applies to the expectation of the loss. In Lemma G.1 we prove theoretically that in expectation, GNSA over a minibatch is equal to GNSA of the whole dataset.

Figure 2 shows that GNSA over a number of trials approximates the GNSA of the entire dataset while RTD fails to do so. Here, we use the output embeddings of two GCN models with the exact same architecture but different initializations trained on node classification on the Cora dataset. The output embedding is the output of the last layer before softmax is applied to get the classification vectors. We compare the convergences with a batch size of 200 and 500.

While LNSA does not exhibit formal convergence under mini-batching, it intuitively demonstrates favorable behavior in such settings. LNSA's design, which relies on nearest-neighbor computations to assess structural preservation within local neighborhoods, ensures that these neighborhoods are typically preserved in expectation. This is because the sampling process in mini-batching generally retains a sufficient number of neighboring points in each batch. By appropriately adjusting the number of nearest neighbors considered in each mini-batch, LNSA is able to approximate the structural consistency of local representations, ensuring that the loss function remains representative of the global structure on a local scale.

4.2.2 Defining NSA-AE: Autoencoder using NSA

To evaluate the efficacy of NSA as a structural discrepancy minimization metric, we take inspiration from TopoAE Moor et al. (2020) and RTD-AE Trofimov et al. (2023), and use NSA as loss function in autoencoders for dimensionality reduction. A normal autoencoder minimizes the MSE loss between the original X and reconstructed embedding $\hat{X}$. We add NSA-Loss as an additional loss term that aims to minimize the discrepancy in representation structure between the original embedding space



(a) (b)

Figure 2: Expectation of subset metrics over a large number of trials. (a) Mean subset GlobalNSA variation over increasing trials. (b) Mean subset RTD variation over increasing trials.

X and the latent embedding space Z. The autoencoder is built as a compression autoencoder where the encoder attempts to reduce the original data to a latent dimension and the decoder attempts to reconstruct the original embedding. Since NSA can be used in autoencoders with mini-batch training, NSA-AE runs almost as fast as a regular autoencoder. We compare the performance of NSA-AE against PCA, UMAP, a regular autoencoder, TopoAE and RTD-AE on four real world datasets.

The performance of NSA-AE is evaluated on the structural and topological similarity between the input data X and the latent data Z. We use the following evaluation metrics: (1) linear correlation of pairwise distances, (2) triplet distance ranking accuracy (Wang et al., 2021), (3) RTD (Barannikov et al., 2022), (4) GNSA, (5) LNSA and (6) kNN-Consistency. As seen in Table 1, NSA-AE has better structural similarity between the original data and the latent data compared to the other models. NSA-AE achieves MSE and running times similar to a normal autoencoder while previous works are several times slower, as presented in Appendix K. We also present the hyperparameters used along with dataset statistics in Appendices L and O.

Table 1: Autoencoder results. L.C = Linear Correlation, T.A. = Triplet distance ranking accuracy, RTD = Representation Topology Divergence, GNSA = GlobalNSA, LNSA@100 = LocalNSA on 100 nearest neighbors, kNN C @ 100 = 100 nearest neighbors consistency. NSA-AE outperforms or almost matches all other approaches on all the evaluation metrics. RTD-AE, which explicitly minimizes on RTD occasionally has lower LID and kNN-C.

		Quality measure					
Dataset	Method	L. C. ↑	T. A. ↑	RTD ↓	GNSA ↓	LNSA @ 100 ↓	kNN-C @ 100 ↑
MNIST	PCA	0.910	0.871 ± 0.008	6.69 ± 0.21	0.0817 ± 0.0025	0.0267 ± 0.00	0.584
	UMAP	0.424	0.620 ± 0.013	18.06 ± 0.48	0.2305 ± 0.0031	1.0353 ± 0.02	0.353
	AE	0.801	0.778 ± 0.007	7.47 ± 0.20	0.0571 ± 0.0011	0.0529 ± 0.00	0.619
	TopoAE	0.765	0.771 ± 0.010	6.16 ± 0.23	0.0477 ± 0.0011	0.0335 ± 0.00	0.631
	RTD-AE	0.837	0.811 ± 0.004	4.26 ± 0.14	0.1694 ± 0.0024	0.0133 ± 0.00	**0.709**
	NSA-AE	**0.947**	$\mathbf{0.886 \pm 0.005}$	$\mathbf{4.29 \pm 0.11}$	$\mathbf{0.0222 \pm 0.00}$	$\mathbf{0.0123 \pm 0.00}$	0.634
F-MNIST	PCA	0.978	$\mathbf{0.951 \pm 0.006}$	5.91 ± 0.12	0.1722 ± 0.0038	0.0566 ± 0.00	0.513
	UMAP	0.592	0.734 ± 0.012	12.16 ± 0.39	0.1420 ± 0.0011	0.6738 ± 0.01	0.365
	AE	0.872	0.850 ± 0.008	5.60 ± 0.21	0.0527 ± 0.0028	0.0515 ± 0.00	0.614
	TopoAE	0.875	0.854 ± 0.009	4.27 ± 0.15	0.111 ± 0.0027	0.0291 ± 0.00	0.601
	RTD-AE	0.949	0.902 ± 0.004	3.05 ± 0.12	0.0349 ± 0.0015	$\mathbf{0.0104 \pm 0.00}$	**0.661**
	NSA-AE	**0.985**	0.939 ± 0.003	$\mathbf{2.78 \pm 0.09}$	$\mathbf{0.0114 \pm 0.00}$	0.0133 ± 0.00	0.633
CIFAR-10	PCA	0.972	0.926 ± 0.009	4.99 ± 0.16	0.1809 ± 0.0046	0.0175 ± 0.00	0.432
	UMAP	0.756	0.786 ± 0.010	12.21 ± 0.22	0.1316 ± 0.0026	0.2880 ± 0.00	0.119
	AE	0.834	0.836 ± 0.006	4.07 ± 0.28	0.0616 ± 0.0019	0.0224 ± 0.00	0.349
	TopoAE	0.889	0.854 ± 0.007	3.89 ± 0.11	0.0625 ± 0.0014	0.0277 ± 0.00	0.360
	RTD-AE	0.971	0.922 ± 0.002	$\mathbf{2.95 \pm 0.08}$	0.0113 ± 0.0003	0.0124 ± 0.00	0.416
	NSA-AE	**0.985**	$\mathbf{0.941 \pm 0.003}$	3.04 ± 0.16	$\mathbf{0.0086 \pm 0.00}$	$\mathbf{0.0103 \pm 0.00}$	**0.437**
COIL-20	PCA	0.966	$\mathbf{0.932 \pm 0.005}$	6.49 ± 0.23	0.2204 ± 0.0	0.2000 ± 0.00	0.801
	UMAP	0.274	0.567 ± 0.016	15.50 ± 0.67	0.1104 ± 0.0	3.3023 ± 0.00	0.569
	AE	0.850	0.836 ± 0.008	9.57 ± 0.27	0.0758 ± 0.0	0.5678 ± 0.00	0.683
	TopoAE	0.804	0.805 ± 0.011	7.33 ± 0.21	0.0676 ± 0.0	0.1044 ± 0.00	0.654
	RTD-AE	0.908	0.871 ± 0.005	5.89 ± 0.10	0.0523 ± 0.0	0.0496 ± 0.00	0.761
	NSA-AE	**0.976**	0.919 ± 0.005	$\mathbf{3.55 \pm 0.18}$	$\mathbf{0.0139 \pm 0.00}$	$\mathbf{0.0085 \pm 0.00}$	**0.809**

4.3 Downstream Task Analysis on Link Prediction

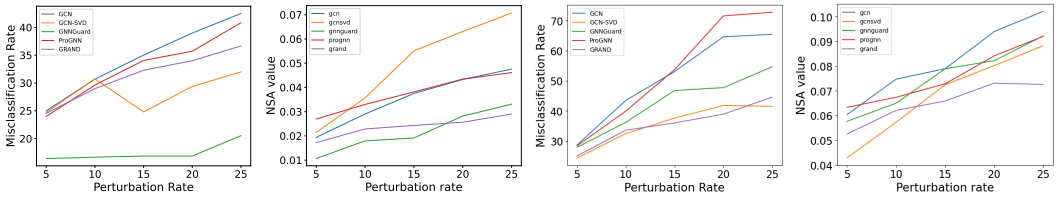
We demonstrate that NSA-AE effectively preserves the structural integrity between the original data and the latent space. However, it is imperative to note that such preservation does not inherently ensure the utility of the resultant latent embeddings. Metrics like NSA, which focus on exact structure preservation, excel in scenarios where the preservation of semantic relationships between data points holds paramount importance.

Link prediction is an ideal task to demonstrate NSA's structure preserving ability. Successful link prediction mandates a global consistency within the embedding space, necessitating that nodes with likely connections are proximate while dissimilar nodes are distant. Any form of clustering or localized alterations to the representation structure can disrupt node interrelationships. In our study, we employ a Graph Convolutional Network (GCN) to train link prediction models across four distinct graph datasets. The original GCN embedding space encompasses 256 dimensions. This space is subsequently processed through NSA-AE, resulting in a reduced-dimensional representation. The latent embeddings are then tested on their ability to predict the existence of links between nodes. Notably, the latent embeddings produced by NSA-AE exhibit superior performance when compared to both a conventional autoencoder and RTD-AE as shown in Table 2. Additionally, we demonstrate that NSA can work at scale by working with the Word2Vec dataset. We present results showcasing the effectiveness of semantic textual similarity matching using Word2Vec embeddings in the Appendix J.

Table 2: Downstream task analysis with Link Prediction. The output embeddings from a GCN trained on link prediction are utilized. Latent embeddings are obtained at various dimensions and ROC-AUC scores are calculated on the latent embeddings obtained after reducing the dimensionality of the original output embeddings. NSA-AE outperforms both a regular autoencoder and RTD-AE across almost all datasets and all latent dimensions.

		Dataset			
Method	Latent Dim	Amazon Comp	Cora	Citeseer	Pubmed
AE	128	50.478	50.0	50.0	51.17
	64	50.56	50.0	50.0	50.03
	32	59.63	49.84	49.80	52.78
RTD-AE	128	94.63	86.52	90.54	91.80
	64	91.63	82.38	89.23	**97.26**
	32	88.34	83.13	57.18	87.05
NSA-AE	128	**95.75**	**97.83**	**97.91**	**97.37**
	64	**94.60**	**97.89**	**98.38**	96.97
	32	**95.76**	**97.68**	**96.11**	**97.66**

4.4 Analyzing Adversarial Attacks



(a) (b) (c) (d)

Figure 3: Robustness tests on GNN architectures with NSA. (a) Misclassification Rate against Data Perturbation Rate under global evasion attack. (b) NSA against perturbation rate under global evasion attack. (c) Misclassification Rate against Data Perturbation Rate under global poisoning attack. (d) NSA against perturbation rate under global poisoning attack

Adversarial attacks have become a critical area of concern across various domains in machine learning, from computer vision to natural language processing and graph-based data. These attacks exploit vulnerabilities in model architectures, to introduce subtle perturbations in the model's representation space to drastically alter its predictions. A structural discrepancy analysis term like NSA can help analyze and identify vulnerabilities in these networks. By comparing the underlying structure of data representations before and after an attack, NSA allows us to quantitatively assess how adversarial